

A Packing of 8 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 8 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.539491. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.499762, obtained by deleting 1 of the 9 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 8 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.539491 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 8$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.499762, obtained by deleting 1 of the 9 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 8/7 = 2.539683$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N8.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
253949199454638043981057155682384852469748326372592443806563627490312955
008029103291081421973526411309823009091399161273223427525458093752659329
0851366028322063092595428842658043054860110769516
```

denominator:

```
999999999999475588189994986664085786952375309093354341920478326839431140
223082263712284875215279458281205216453753215986054306966140494653472553
037028382256123130741947814192369421466354264945
```

B Coordinates

8 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N8.txt`.

```
-364111935698 617597518371 244640932906 28428715566662878345965 764126962371 -92409251591 -405144561696452755632 -77099606934 49107774090 -432928189736
-620874451974 -152121766023 -179946989313 -326994456327108391842 -89234558454 497859085742 2045029925266584774593 145674904381 360791965168 365597632849
-58205066430 96641728303 13514613787 200386124532 -229028813211 -78061240455 -457685078107 -112157834833010191785 367418710315 -203417051196 -146384221924
136921744183 -281481941138 -105797991206 34114293863323894304 -94308709991 -688125718715 191662406467662097995766 406209258121 -81233187596 155913094798
478638505522 11808569214 -51279168985 -5465312212279353209522 119405677550 568469039449 260038596757 -97644242693 224515043969 335498393764 -592506215424
75744317151 -331143670696 604889010425 8167554546 -214645804756 -472521594760 32726274577 36483478989422699993138 -724273377236 -161258211962 -41092508142
104480396256 -13733331268 667303783454 46789757530 -391890902470 601884054066 -77228539628 444034814166120310121739 -633110043301 61394859198 297141439828
458040554869 484118231092 -183242115115 18469845439666250184768 -412869383453 -82608202687 4308986680550179576399 -65093352109 252050318445 652899576697
-65212696041 403120239824 221363400286 -5662653529353054935463 -417790102653 343665395818 7926545319235082487241 571155191961 146818639471 474438971060
253664689566 380427905299 -20996983991 -21685465117640006166816 -25836827665 167113606521 -4334452802814688897541 91747315406 240052744306 -348122180530
273930339678 -83694155510 438990118604 269739626365-312795593286 -198200237684 -206980698938 -40363268399328743152 -197032922266 564971270084 -9524815630
-440250281999 -166510244566 -600418668889 1407451077338753417662 79664959589 24345500602 150410360085 -262092721823 10679443743 555485502013 -210538753255
-418110925938 64217925704 398117342846 -17782448922439304742983 -109809065918 327887499524 4233978239555800950627 337702791736 -160868407802 -92424724401
255805220561 131630693122 -420104624825 -70454320838191943663679 -352636144735 157169021105 -245198964639890294038 139691372907 -127767200690 287885828335
371201775158 283702279677 57424050838 208409577819 -162069536137 -601841564221 71920077651 10680692231209625159688 -296320480577 -428439603821 57531287804
244135622679 370675074234 -280627837046 -308926106095842641265 27838356622 838968723457 -100483803208-616791402886 -111179641278 390398574605 155084773503
16566898140 -46866913125 -155229699682 7649233212481722131256 121013964685 32233281665 -667772783190 -364353854916 428031863902 156747191856 -387866433128
-449115291363 760927166427 72202117719 315978591118-582341727899 604914761862 303950220267 80096543767273526396214 254882810888 -286200203027 -231684349704
454420501876 476357820025 -222247029713 26142113182491050528495 -137282293058 193542078135 -200252114038381733809 440825167301 -175500068175 533178588683
-165059505722 223577158910 357782055708 29696930192639906798936 -265973058233 99401031505 -35083797104817259366837 26123109223 -322413158241 316663481297
6580640727 53840076220 -404933773327 587011703385 39743643365 174491064197 -134100141807 -3515435955507148450946 -262118595972 340451319288 -12598746281
133924867677 22122611919 63460693673 384552176921 13303658787 289873229897 -247534708837 -1196977041695405238844 404346363952 -311299174535 -158460729956
647407474732 -76663784226 -5563478108 -35843227382458851797931 30451608992 433687544890 453245304881 91268972034 175776322459 225241570162 595677194787
-305437859081 -258912260551 -434384525426 63154130909418990676 113575010143 -656900719360 2462991342055126354729 530427999515 585499918225 -84709418780
```